| RUBRIC COMPONENT #3 (4 MARKS) | OFFICIAL CRITERIA: SQL database implementation - DDL with constraints |
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# SQL Database Implementation — DDL & Indexes

**Course:** DBMS Mini Project (UE24CS395) • **Institution:** PES University (2026–2027)
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## 1. Schema Implementation (database/sql/schema.sql)

The DDL defines all 22 tables with strict primary keys, foreign keys, CHECK constraints, and 17 dedicated performance indexes.

```
-- Core Entity DDL Excerpt (PostgreSQL / SQLite Compatible)
CREATE TABLE officers (
    officer_id VARCHAR(50) PRIMARY KEY,
    full_name VARCHAR(150) NOT NULL,
    badge_number VARCHAR(50) NOT NULL UNIQUE,
    department VARCHAR(100) NOT NULL,
    contact_email VARCHAR(120) NOT NULL UNIQUE,
    role VARCHAR(100) NOT NULL DEFAULT 'INVESTIGATOR' CHECK (role IN ('ADMIN', 'INVESTIGATOR')),
    account_status VARCHAR(50) NOT NULL DEFAULT 'Active' CHECK (account_status IN ('Active', 'Suspended', 'On Leave')),
    hashed_password VARCHAR(255) NOT NULL,
    created_at TIMESTAMP WITH TIME ZONE DEFAULT CURRENT_TIMESTAMP
);

CREATE TABLE cases (
    case_id VARCHAR(50) PRIMARY KEY,
    title VARCHAR(200) NOT NULL,
    description TEXT NOT NULL,
    case_type VARCHAR(100) NOT NULL,
    status VARCHAR(50) NOT NULL DEFAULT 'Open' CHECK (status IN ('Open', 'Under Investigation', 'On Hold', 'Closed')),
    priority VARCHAR(50) NOT NULL DEFAULT 'Medium' CHECK (priority IN ('Low', 'Medium', 'High', 'Critical')),
    lead_officer_id VARCHAR(50) REFERENCES officers(officer_id) ON DELETE SET NULL,
    incident_date TIMESTAMP WITH TIME ZONE DEFAULT CURRENT_TIMESTAMP
);

CREATE TABLE case_people (
    case_id VARCHAR(50) NOT NULL REFERENCES cases(case_id) ON DELETE CASCADE,
    person_id VARCHAR(50) NOT NULL REFERENCES people(person_id) ON DELETE CASCADE,
    role_in_case VARCHAR(100) DEFAULT 'Subject of Investigation',
    added_at TIMESTAMP WITH TIME ZONE DEFAULT CURRENT_TIMESTAMP,
    PRIMARY KEY (case_id, person_id)
);
```

## 2. B-Tree & Compound Performance Indexes (17 Indexes)

| Index Name | Table & Columns | Optimization Objective |
|---|---|---|
| idx_cases_lead_officer | cases(lead_officer_id) | Accelerates officer caseload joins and integrity lookups |
| idx_cases_status_priority | cases(status, priority) | Compound index eliminating full-table scans during dashboard triage |
| idx_evidence_case | evidence(case_id) | Sub-second exhibit retrieval when opening case dossiers |
| idx_custody_timestamp | evidence_custody(event_timestamp DESC) | Chronological ordering of custody handoffs for court reports |
| idx_vehicles_registration | vehicles(registration_number) | Exact lookup for ANPR camera triggers and police dispatch |
| idx_devices_serial | devices(serial_number) | Exact lookup for hardware forensic exhibits and IMEI extractions |
| idx_cp_person | case_people(person_id) | Accelerates recursive CTE associate traversals across shared cases |